Ex:15:- LCM & HCF of $2^8.7^{11}.5^{18}$, $2^{48}.3^{17}.7^3.5^{16}$, $2^6.5^7.7^{45}.3^{68}$?

Ans:- LCM is product of highest powered prime factors

$= 2^{48}.3^{68}.5^{18}.7^{45}$

HCF is product of least powered common prime factors

$= 2^6.5^7.7^3$

Ex:16:- What is unit digit of $(16)^{85} + (763)^{143} + (147)^{88} + (92)^{863}$

Ans:- $(16)^{85} + (763)^{143} + (147)^{88} + (92)^{863}$

$\downarrow$ $\quad$ $143/4 \to R \to 3$ $\quad$ $88/4 \to R \to 0$ $\quad$ $\frac{63}{4} \to 3$

6 $\quad$ $3^3 \to ud \to 7$ $\quad$ $7^4 ud \to 1$ $\quad$ $2^3 ud \to 8$

$= 6+7+1+8 = 22$

Ex:17:- U.d of $(149)^{642} - (784)^{863}$

9 even power $\quad$ 4 odd power

u.d '1' $\quad$ '4'

U.d = $1-4 = -3$ there is no '-ve' unit digit.

It means '3' less to 10 which is '7'.

When '-ve' answer came then simple add '10' & take the result as answer.

U.d = $-3+10 = 7$

Ex:18:- How many times we have to type/Press keyboard buttons to type nos from 1 to 600?

Ans:- to type 1-9 (single digited) $\longrightarrow$ $9 = 9\times1$ (1-9)

" 10-99 (double ") $\longrightarrow$ $180 = 90\times2$ (10-99)

" 100-600 (triple ") $\longrightarrow$ $1503 = 501\times3$ (100-600)

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Booklet Questions:-

Ex:-19-[B.Q:-8]:- Find the least number, which when devided by 12,15,20 & 54 leaves a remainder of 8 in each case?

Ans:- = LCM(12,15,20,54) + 8

$= 2^2 \times 3^3 \times 5^1 + 8$

$= 540 + 8$

$= \boxed{548}$

2	12,15,20,54
2	6,15,10,27
3	3,15,5,27
3	1,5,5,9
3	1,5,5,3
5	1,5,5,1
	1,1,1,1.

Ex:-20:- [B.Q:-9]:- The traffic lights at Silk board crossing change after every 40 sec, 72 sec & 108 sec respectively. If they all changed at 5:20:00 hr's, then find next simultaneous change at which time?

Ans:- While lights are changing, whenever common multiple of all time laps comes @ that time all lights will change simultaneously. ie., LCM of time laps.

LCM (40, 72, 108). = LCM is product of highest powers of prime factors of all numbers

$40 = 8 \times 5 = 2^3 \times 5^1$

$72 = 8 \times 9 = 2^3 \times 3^2$

$108 = 12 \times 9 = 4 \times 3 \times 3^2 = 2^2 \times 3^3$

$= 2^3 \times 3^3 \times 5^1 = 1080$ sec's

For every 1080 seconds all lights changes simeltaneo-sly. 1080 sec in mins $= \frac{1080}{60} = 18$ mins

Ans:- 5:20:00 + 18 mins = $\boxed{5:38:00}$ hr's

Ex:-21:-[B·Q:-10]:- A number when devided by a divisor leaves a remainder of 24. When twice the original number is devided by the same divisor, the rem is '11'. What is the value of the divisor?

Ans:- Lets assume the divisor as 'x'.

dividend in 1st case = D = $x \times y + 24$.

double of D = 2D = $2xy + 48$

when 2D devided by x gives '11' rem

$\Rightarrow \frac{2D}{x} \to R = 11 \Rightarrow \frac{2xy+48}{x} \Rightarrow \frac{2xy}{x} \to R \to 0 + \frac{48}{x} \to R \to 11$

$\Rightarrow \frac{48}{x} \to R = 11$ $\Rightarrow$ means if $x = 37$ then only it is right

$\Rightarrow \therefore \boxed{x = 37}$

Ex:-22:-[B·Q:-11]:- The GCD & LCM of two no·s are 66 & 384. If one of the nos devided by 2 it results 192, what is the 2nd number.

Ans:- 1st number. x = dividend = divisor × Quotient

$= 2 \times 192 = 384$

$\Rightarrow$ LCM × HCF = Product of 2 no·s

$66 \times 384 = 384 \times N_2 \Rightarrow \boxed{N_2 = 66}$

Ex:-23:-[B·Q:-2]:- Find the remainder when

$1! + 2! + 3! + 4! + 5! + \dots + 1000!$ is devided by 8?

Ans:- $\frac{1! + 2! + 3! + 4! + 5! + \dots + 1000!}{8} \to R = ?$

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Ans:- $\frac{f(x+a_1)+f(x+a_2)+f(x+a_3)+\cdots}{f(x)}$ Rem Qs

$= \frac{f(x+a_1)}{f(x)} + \frac{f(x+a_2)}{f(x)} + \cdots \Rightarrow$ Rem Qs $f(a_1)+f(a_2)+\cdots$

if $f(a_1)+f(a_2)+\cdots > f(x)$ continue till to get $f(a_1)+f(a_2)+\cdots$ Qs $< f(x)$ [then only remainder]

$\frac{1!}{8} + \frac{2!}{8} + \frac{3!}{8} + \frac{4!}{8} + \frac{5!}{8} + \frac{6!}{8} + \cdots \frac{1000!}{8}$

Rem $= 1+2+6+0+0+0+\cdots 0 > 8$

$\Rightarrow$ Rem $= \frac{1+2+6}{8} = \frac{9}{8} =$ [1]

Ex-24:- Find the greatest number which will devide 410, 751 & 1030 leaves remainder 7 in each case Qs.

Ans:- HCF $(410-7, 751-7, 1030-7)$

$=$ HCF $(403, 744, 1023)$

$= 31$

403 | 744 | 1
403

341 | 403 | 1
341

62 | 341 | 5
310

31 | 62 | 2
62

0

31 | 1023 | 33
1023

0